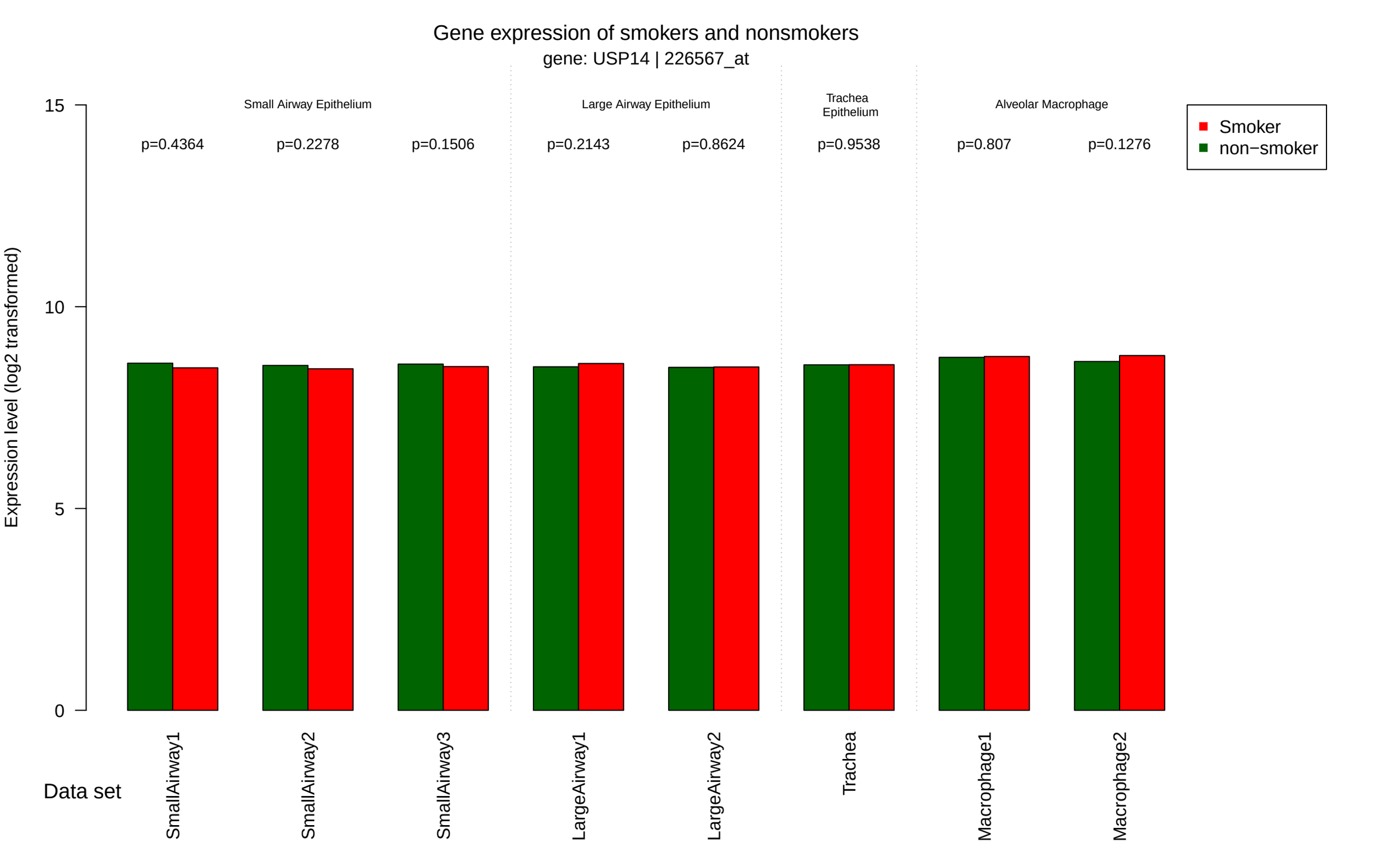
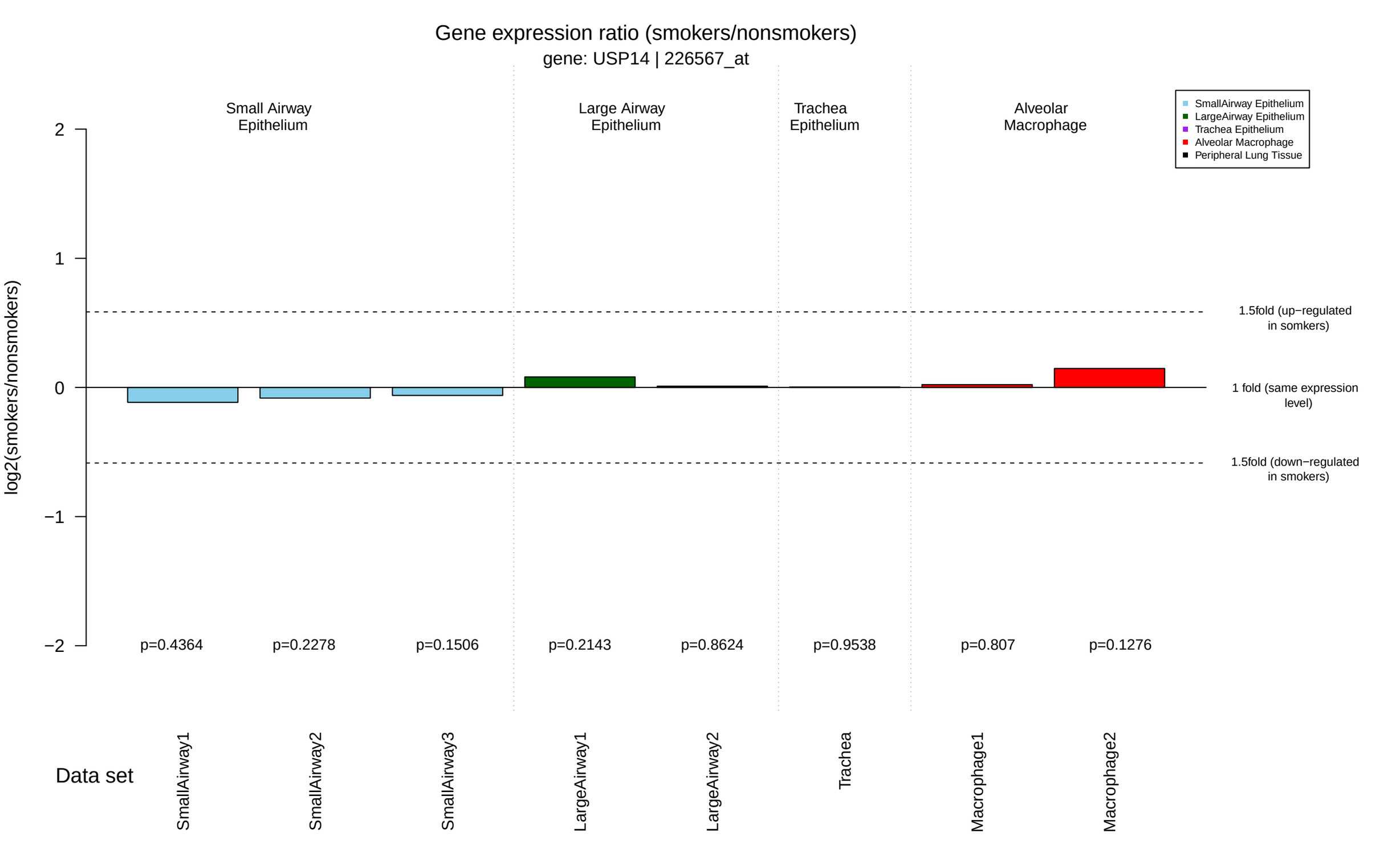
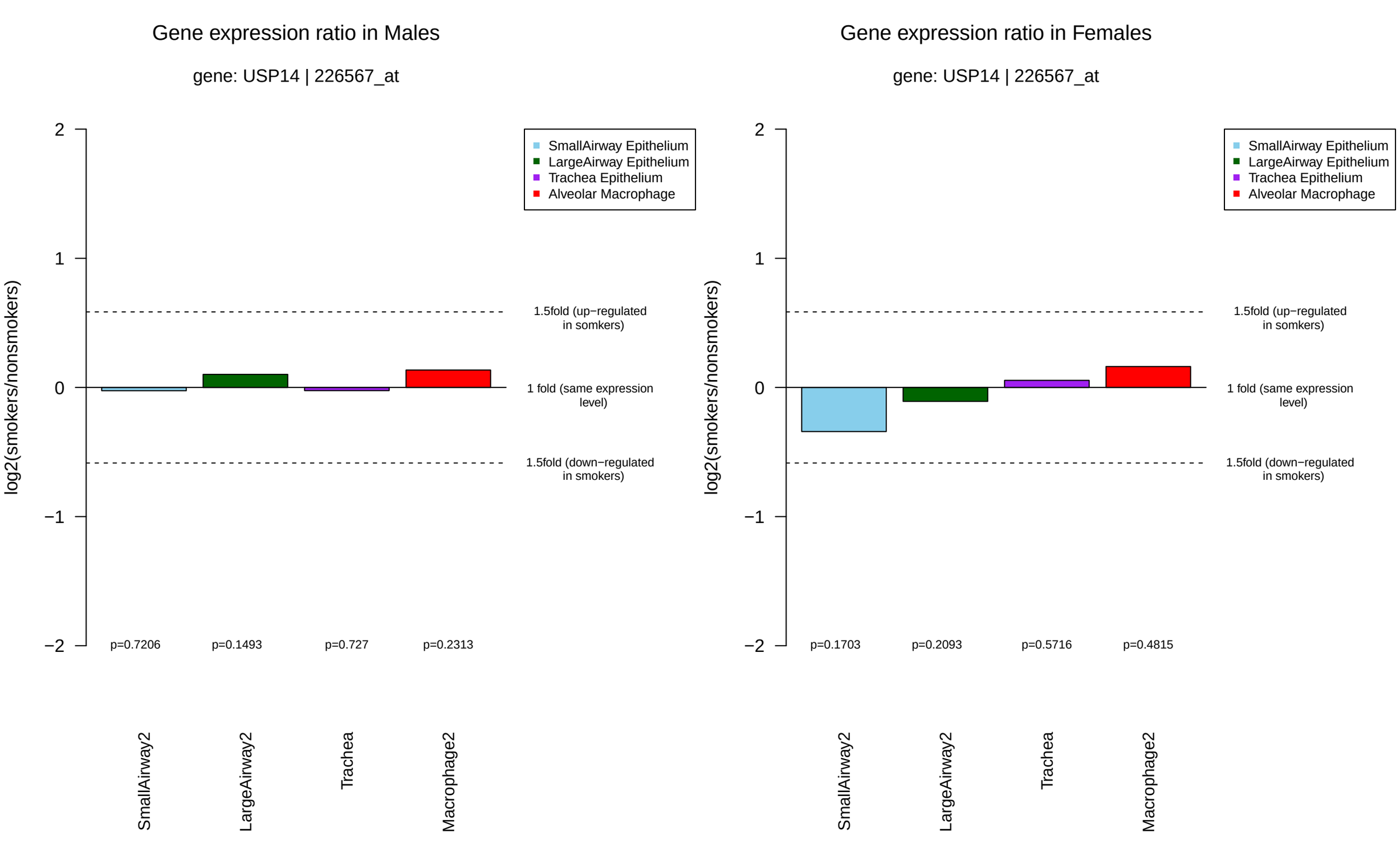
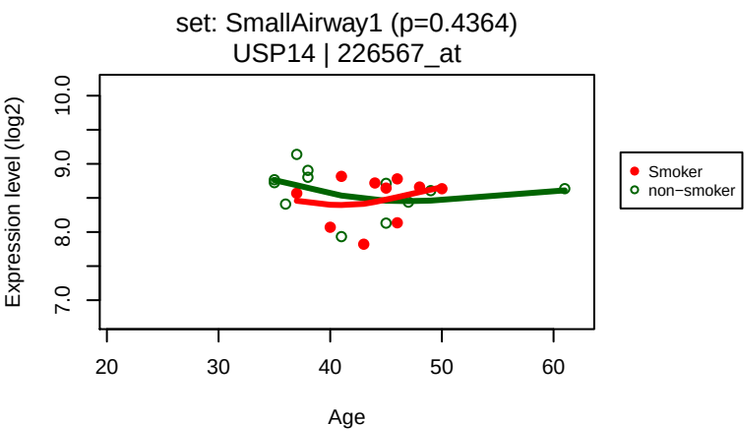
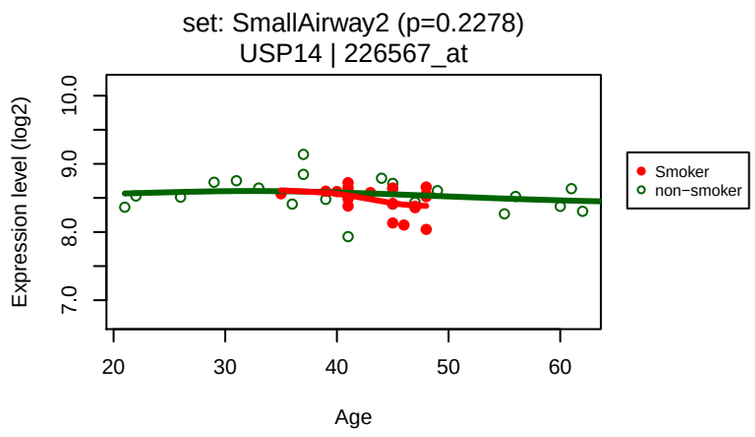
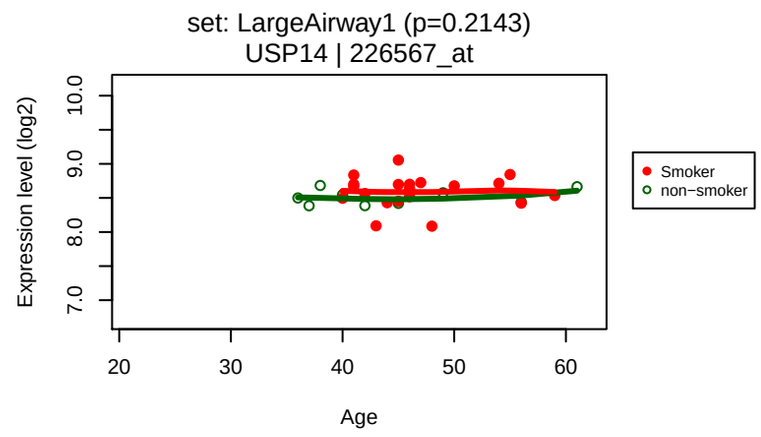

Gene expression of smokers and nonsmokers
gene: USP14 | 226567_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4364
p=0.2278
p=0.1506
p=0.2143
p=0.8624
p=0.9538
p=0.807
p=0.1276
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

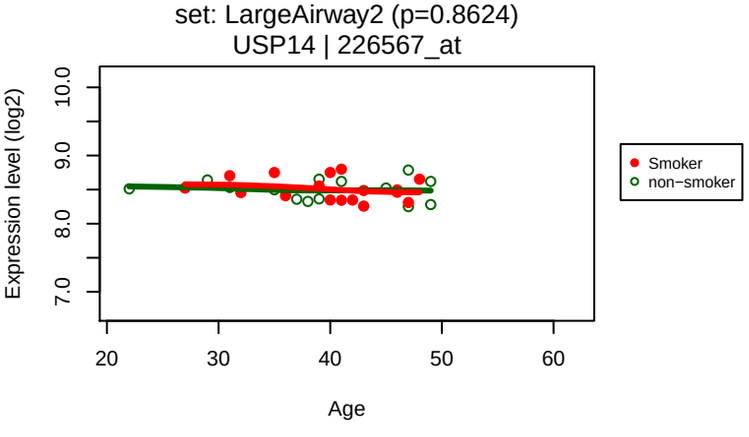

Gene expression ratio (smokers/nonsmokers)
gene: USP14 | 226567_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4364
p=0.2278
p=0.1506
p=0.2143
p=0.8624
p=0.9538
p=0.807
p=0.1276
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

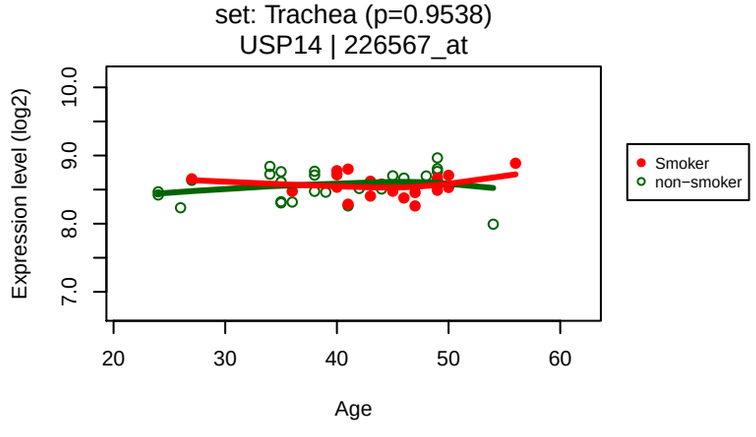

Gene expression ratio in Males
gene: USP14 | 226567_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7206
p=0.1493
p=0.727
p=0.2313
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: USP14 | 226567_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1703
p=0.2093
p=0.5716
p=0.4815
SmallAirway2
LargeAirway2
Trachea
Macrophage2

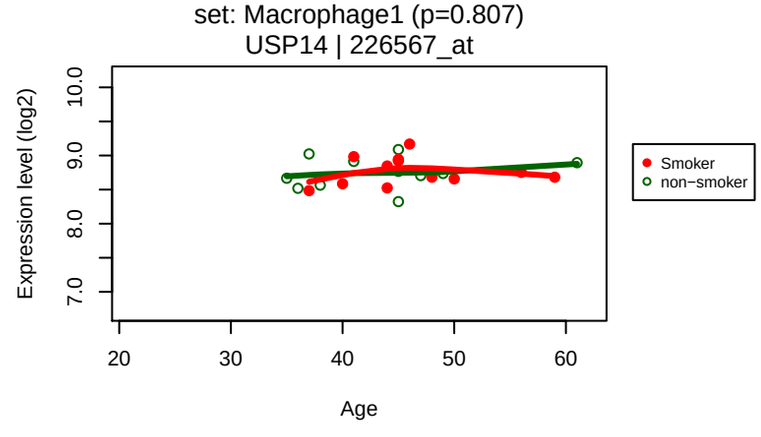

set: SmallAirway1 (p=0.4364)
USP14 | 226567_at
Expression level (log2)
Age
Smoker
non−smoker

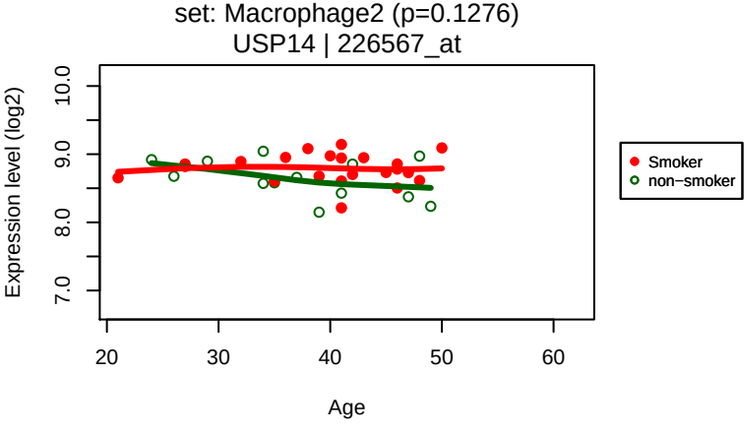

set: SmallAirway2 (p=0.2278)
USP14 | 226567_at
Expression level (log2)
Age
Smoker
non−smoker

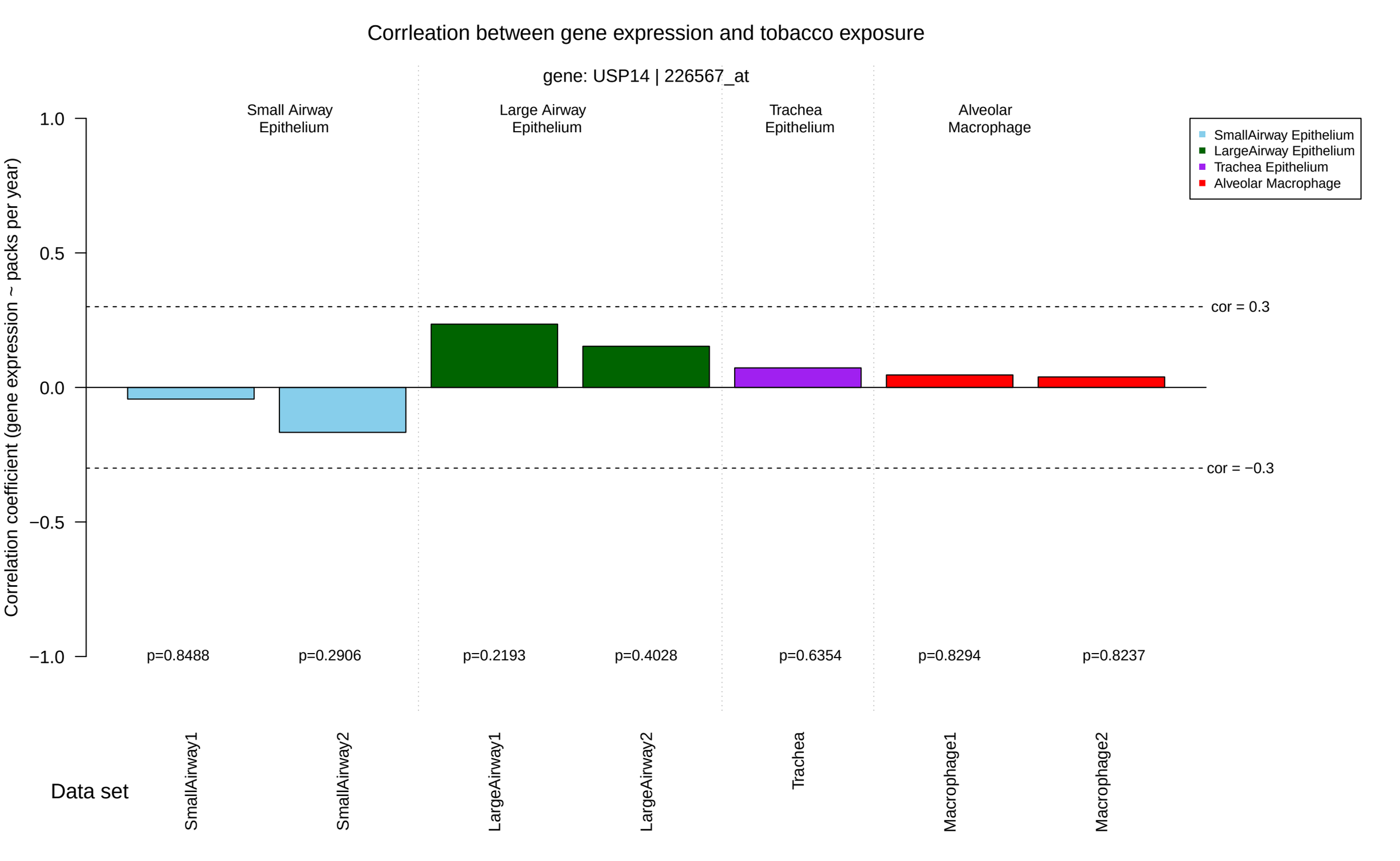

set: LargeAirway1 (p=0.2143)
USP14 | 226567_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8624)
USP14 | 226567_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9538)
USP14 | 226567_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.807)
USP14 | 226567_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1276)
USP14 | 226567_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: USP14 | 226567_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8488
p=0.2906
p=0.2193
p=0.4028
p=0.6354
p=0.8294
p=0.8237
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2